WORKED SOLUTION

STEP II 2023, Question 8

Pure • Vectors in Three Dimensions

STEP II 2023, Question 8

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2023 STEP papers (a zip file):

www.ocr.org.uk/Images/707102-2023-past-papers.zip

The key idea

Name the three pairs of opposite edges, and part (i) becomes three linear equations. After that, everything follows from turning each length condition into a dot product, using $|\mathbf{u}-\mathbf{v}|^2 = |\mathbf{u}|^2 + |\mathbf{v}|^2 - 2\,\mathbf{u}\cdot\mathbf{v}$. Once all three dot products $\mathbf{a}\cdot\mathbf{b}$, $\mathbf{b}\cdot\mathbf{c}$, $\mathbf{c}\cdot\mathbf{a}$ are written in terms of $|\mathbf{a}|^2$, $|\mathbf{b}|^2$ and $|\mathbf{c}|^2$, parts (iii) and (iv) are just expanding brackets.

Solution

Part (i)

Label the tetrahedron $OABC$. Its six edges form three pairs of edges that don't share a vertex:

$$OA \text{ and } BC, \qquad OB \text{ and } CA, \qquad OC \text{ and } AB.$$

Write $OA = p$, $BC = p'$, $OB = q$, $CA = q'$, $OC = r$, $AB = r'$. The four faces have perimeters

$$OAB : p+q+r', \qquad OAC : p+r+q',$$
$$OBC : q+r+p', \qquad ABC : p'+q'+r'.$$

If the tetrahedron is isosceles, then $p = p'$, $q = q'$, $r = r'$, and each of the four perimeters equals $p+q+r$.

Conversely, suppose all four perimeters are equal. Put $X = p-p'$, $Y = q-q'$, $Z = r-r'$. Comparing each of the first three faces with ABC:

$$(p+q+r') - (p'+q'+r') = X+Y = 0,$$
$$(p+r+q') - (p'+q'+r') = X+Z = 0,$$
$$(q+r+p') - (p'+q'+r') = Y+Z = 0.$$

Adding all three gives $2(X+Y+Z) = 0$. Subtracting each equation in turn from $X+Y+Z = 0$ gives $Z = 0$, $Y = 0$ and $X = 0$. So $p = p'$, $q = q'$ and $r = r'$: the tetrahedron is isosceles. That proves both directions.

Part (ii)

Since $\overrightarrow{BC} = \mathbf{c} - \mathbf{b}$, the condition $OA = BC$ says

$$|\mathbf{a}|^2 = |\mathbf{c} - \mathbf{b}|^2 = |\mathbf{b}|^2 + |\mathbf{c}|^2 - 2\,\mathbf{b} \cdot \mathbf{c},$$

which rearranges to $2\,\mathbf{b} \cdot \mathbf{c} = |\mathbf{b}|^2 + |\mathbf{c}|^2 - |\mathbf{a}|^2$, as required.

The same argument with the other two pairs of opposite edges ($OB = CA$ and $OC = AB$) gives

$$2\,\mathbf{a} \cdot \mathbf{c} = |\mathbf{a}|^2 + |\mathbf{c}|^2 - |\mathbf{b}|^2, \qquad 2\,\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}|^2 + |\mathbf{b}|^2 - |\mathbf{c}|^2.$$

Adding these two, the $|\mathbf{b}|^2$ and $|\mathbf{c}|^2$ terms cancel:

$$2\,\mathbf{a} \cdot \mathbf{b} + 2\,\mathbf{a} \cdot \mathbf{c} = 2|\mathbf{a}|^2, \quad \text{so} \quad \mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = |\mathbf{a}|^2.$$

The three dot-product formulas are symmetric in $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$, so in the same way

$$\mathbf{b} \cdot (\mathbf{c} + \mathbf{a}) = |\mathbf{b}|^2, \qquad \mathbf{c} \cdot (\mathbf{a} + \mathbf{b}) = |\mathbf{c}|^2.$$

Part (iii)

Let $S = |\mathbf{a}|^2 + |\mathbf{b}|^2 + |\mathbf{c}|^2$. Adding the three formulas for $2\,\mathbf{b} \cdot \mathbf{c}$, $2\,\mathbf{c} \cdot \mathbf{a}$ and $2\,\mathbf{a} \cdot \mathbf{b}$ from part (ii), each squared length appears twice with a plus sign and once with a minus sign, so

$$2(\mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{c} + \mathbf{c} \cdot \mathbf{a}) = S.$$

Distance from O.

$$OG^2 = \tfrac{1}{16}|\mathbf{a} + \mathbf{b} + \mathbf{c}|^2 = \tfrac{1}{16}(S + 2(\mathbf{a} \cdot \mathbf{b} + \mathbf{b} \cdot \mathbf{c} + \mathbf{c} \cdot \mathbf{a})) = \tfrac{1}{16}(2S) = \tfrac{1}{8}S.$$

Distance from A. Here $\overrightarrow{AG} = \tfrac{1}{4}(\mathbf{b} + \mathbf{c} - 3\mathbf{a})$, and

$$\begin{aligned} |\mathbf{b} + \mathbf{c} - 3\mathbf{a}|^2 &= |\mathbf{b} + \mathbf{c}|^2 - 6\,\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) + 9|\mathbf{a}|^2 \\ &= |\mathbf{b}|^2 + |\mathbf{c}|^2 + 2\,\mathbf{b} \cdot \mathbf{c} - 6|\mathbf{a}|^2 + 9|\mathbf{a}|^2 \qquad \text{using } \mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = |\mathbf{a}|^2 \\ &= |\mathbf{b}|^2 + |\mathbf{c}|^2 + (|\mathbf{b}|^2 + |\mathbf{c}|^2 - |\mathbf{a}|^2) + 3|\mathbf{a}|^2 \\ &= 2S. \end{aligned}$$

So $AG^2 = \tfrac{1}{16}(2S) = \tfrac{1}{8}S$ too.

Distances from B and C. Every fact we used about $\mathbf{a}$ holds equally for $\mathbf{b}$ and for $\mathbf{c}$ (the end of part (ii)), and S is symmetric. So the identical calculation gives $BG^2 = CG^2 = \tfrac{1}{8}S$.

Hence G is at distance $\sqrt{S/8}$ from all four vertices.

Part (iv)

Expand the suggested vector:

$$\begin{aligned} |\mathbf{a} - \mathbf{b} - \mathbf{c}|^2 &= S - 2\,\mathbf{a} \cdot \mathbf{b} - 2\,\mathbf{a} \cdot \mathbf{c} + 2\,\mathbf{b} \cdot \mathbf{c} \\ &= S - 2|\mathbf{a}|^2 + 2\,\mathbf{b} \cdot \mathbf{c} \qquad \text{by part (ii)} \\ &= (|\mathbf{b}|^2 + |\mathbf{c}|^2 - |\mathbf{a}|^2) + 2\,\mathbf{b} \cdot \mathbf{c} = 4\,\mathbf{b} \cdot \mathbf{c}. \end{aligned}$$

So

$$\mathbf{b}\cdot\mathbf{c} = \tfrac{1}{4}|\mathbf{a}-\mathbf{b}-\mathbf{c}|^2 \geq 0,$$

and the angle BOC is not obtuse.

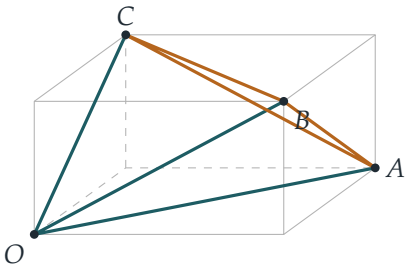
Ask yourself That is one angle out of twelve. How do we get all the others without repeating the work?

The same calculation with $\mathbf{b}-\mathbf{c}-\mathbf{a}$ and $\mathbf{c}-\mathbf{a}-\mathbf{b}$ gives $\mathbf{c}\cdot\mathbf{a}\geq 0$ and $\mathbf{a}\cdot\mathbf{b}\geq 0$, so no angle at O is obtuse. Being isosceles is a property of the tetrahedron, not of how we label it, so any vertex could have been called O. The same argument therefore works at A, B and C. (Alternatively, every face has sides p, q, r, so all four faces are congruent triangles. Their angles are exactly the three angles BOC, COA, AOB at O.) Hence none of the angles between edges meeting at a vertex is obtuse.

Right angles? Equality $\mathbf{b}\cdot\mathbf{c}=0$ happens only if $|\mathbf{a}-\mathbf{b}-\mathbf{c}|=0$, that is $\mathbf{a}=\mathbf{b}+\mathbf{c}$. But then A lies in the plane through O, B, C (in fact $OBAC$ is a rectangle), and $OABC$ is flat, not a tetrahedron. The same holds at every angle, so **no angle can be a right angle**: in a genuine isosceles tetrahedron every face angle is acute.

Checks

- A concrete example: take a cuboid with side lengths x,y,z and pick four of its vertices, no two joined by an edge of the cuboid, e.g. $O=(0,0,0)$, $A=(x,y,0)$, $B=(x,0,z)$, $C=(0,y,z)$. Opposite edges are diagonals of opposite faces, so they're equal. With $x=1$, $y=2$, $z=3$ we get $\mathbf{b}\cdot\mathbf{c}=9$ and $\mathbf{a}-\mathbf{b}-\mathbf{c}=(0,0,-6)$, and indeed $\frac{1}{4}\cdot 36=9$. Also $G=(\frac{1}{2},1,\frac{3}{2})$ is the centre of the cuboid, at distance $\sqrt{14}/2$ from every vertex, which matches $\sqrt{S/8}$ with $S=5+10+13=28$.
- The flat case: $\mathbf{b}=(3,0,0)$, $\mathbf{c}=(0,4,0)$, $\mathbf{a}=(3,4,0)$ gives a rectangle whose opposite "edges" (sides and diagonals) are equal and which has a right angle. That is exactly the degenerate case we excluded.
- These identities were also confirmed by computer on thousands of random cuboid-based examples.



Common slips

- In part (i), proving only one direction. "If and only if" needs both, and the converse is the part with content.
- In part (iv), forgetting to rule out the right angle properly. $\mathbf{b} \cdot \mathbf{c} = 0$ forces $\mathbf{a} = \mathbf{b} + \mathbf{c}$, which makes the four points coplanar.
- Checking only the angles at O and stopping. Say why the same holds at the other three vertices.

Beyond STEP

The cuboid picture above is the whole story: every isosceles tetrahedron (also called a *disphenoid*) is obtained by choosing alternate vertices of a cuboid. Then part (iii) says the centroid is the cuboid's centre, and part (iv) says the cuboid's sides have positive length. Converting length conditions into dot products, and so into a *Gram matrix* of inner products, is a standard tool in first-year linear algebra and geometry. Strang's *Introduction to Linear Algebra* is a good place to see it developed.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.